

Seat No.	
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Set **P**

**B.Sc. (E.C.S.) (Semester - I) (New) (NEP CBCS) Examination:
March/April - 2025
Digital Electronics and Microprocessors (G06-0103)**

Day & Date: Thursday, 08-May-2025
Time: 12:00 PM To 01:30 PM

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose correct alternative.**06**

- 1) _____ is IC number of Not gate.

a) 7432	b) 7408
c) 7404	d) 7486
- 2) 8085 is _____ bit microprocessor control signals.

a) 8	b) 16
c) 32	d) 64
- 3) _____ IC is used as counter.

a) 74266	b) 7474
c) 7495	d) 7490
- 4) 8085 has _____ bit program counter.

a) 8	b) 16
c) 20	d) 24
- 5) 4:1 multiplexer has _____ control signals.

a) 2	b) 4
c) 8	d) 14
- 6) ADI has _____ addressing mode.

a) direct	b) indirect
c) immediate	d) register

Q.2 Answer the following. (any Three)**06**

- a) Define synchronous counter.
- b) Explain NOR gate with symbol.
- c) Explain INTR and HOLD control signal.
- d) Draw block diagram of half subtractor.
- e) Define instruction cycle.

- Q.3 Answer the following questions. (Any Two) 06**
- a) Explain Derived gate - XOR, XNOR.
 - b) Explain arithmetic instruction.
 - c) Explain RS flip flop using NAND gate.
- Q.4 Answer the following questions. (Any Two) 06**
- a) Explain 1:4 Demultiplexer.
 - b) Explain addressing modes of 8085 microprocessor.
 - c) Explain GPR's of 8085 microprocessors.
- Q.5 Answer the following questions. (Any One) 06**
- a) Explain asynchronous 3 bit up and down counter.
 - b) Explain data transfer and branch control instructions of 8085 microprocessor.